Effectiveness of Lead Exposure Regulations on US Kidney Health

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Introduction

Since the 1970s, the United States Government has made substantial progress regarding the regulation to limit daily exposure to lead, which had previously been remarkably high compared to other developed countries. However, it should be noted that the EPA has identified numerous violations of these regulations and many firms have been charged as a result. It is unclear to what extent exposure leads due to failure to comply on the part of firms, contractors, municipal governments, and homeowners might result in the US population still being known to have relatively high blood lead levels. This paper seeks to identify whether some age and economic demographics are more affected by lead exposure and its downstream effects than others.

Methods

The 2005-2006 edition of the National Health and Nutrition Examination Survey was collected so that a significant proportion of the study population would simultaneously contain adults who were exposed to lead throughout childhood, adults whose early childhoods occurred after the onset of the 1970s regulations, and allow for a significant amount of time after allow the assessment of lagged mortality rates. Demographic variables, poverty income ratio, BMI, waist circumference, laboratory tests for blood lead, hemoglobin, serum creatinine, diabetes, smoking, cancer statuses and mortality statuses from the 2019 follow-up study were included in the collection.

It was determined that the 1978 CPSC Residential Lead Paint Ban, 1986 Safe Drinking Water Act, and 1992 Residential Lead-Based Paint Hazard Reduction Act were the main regulations of interest, so the population was split between those born pre-1978, between 1978-1988 and which corresponded to those older than 28, between 14 to 18 during the time of the survey. eGFR was computed using the CKD-EPI 2021 equation. Those below the age of 18 at the time were not included due to the fact their laboratory tests are not comparable to adults and the fact that the majority of those under 18 at the time were not included in the mortality follow up due to non-participation and often, ineligibility to participate due to their age. As a result, approximately 60% and 40% of the samples were divided between the pre-1978 and post-1978 born participants.

The resulting data was analyzed for overall response completion and missingness mechanisms. It was determined that depression score, smoking status, and general health status were the most commonly missing features in the data. The overall completion rate was approximately 6.43%. A Little's MCAR test was performed ($p = 0$) and was determined that to a very high degree of confidence, the missing responses were not missing at random. Three common patterns were identified which were usually linked to depression questionnaires and smoking statuses. Hot Deck Imputation was performed as this method has been bench marked to work well on survey data but relies on at least some people of similar health responding honestly to the survey, which could potentially bias the imputation toward making the participants seem less healthy or healthier than they are in reality, which the latter is the more likely.

Correlation tests were performed on the data using both Pearson and Kendall correlations to determine which variables in the data had clear linear relationships with special interest in the age, blood lead concentration, and metabolic markers. It was determined the only clear correlations in the data were BMI being positively correlated with waist circumference and age being negatively correlated with eGFR and serum creatinine, which all were suspected. These patterns were consistent between both the Pearson and Kendall tests. It was noteworthy that blood lead concentration was weakly negatively correlated with eGFR, which could be indicative of lead induced renal damage in some subsection of the sample.

Permutation tests at 10,000 permutations for differences in medians across the two age strata for blood lead concentration ($p = 0$), blood hemoglobin concentration ($p = 0.0$), and eGFR ($p = 0$) resulted in significant results for differences across. Tests were also performed for differences in the mean between strata for which the results were as follows, blood lead concentration ($p = 0$), blood hemoglobin concentration ($p = 0.0$), and eGFR ($p = 0$). These results are consistent with the central tendency of hemoglobin being similar across the two age strata.

When examining the overall distribution, while the differences in central tendency between the two groups for these variables, the other characteristics of the distribution include the skew and dispersion for each. For example, in blood hemoglobin concentration, there are clear differences across the age strata, where the population born before 1978 had a wider variability than the adults born after 1977. Those born after 1977 also had a far wider variability for blood lead concentration than those born before but also had a lower median blood lead concentration. This is likely a case of survivorship bias as tested later where adults with higher lead exposure might face medical intervention or early mortality which may result in there being less adults with extremely high lead exposure. A disproportionately high proportion of adults with elevated blood lead (greater than 3.5 ug/dl) concentrations were no longer alive at the time of the 2019 follow-up study. This was confirmed in Mantel-Haenszel tests which were separately performed controlling for age ($p = 1.104e - 07, \chi^2 = 28.182, OR = 1.675154$), race ($p < 2.2e - 16, \chi^2 = 183.27, OR = 2.827388$), and sex ($p < 2.2e - 16, \chi^2 = 176.66, OR = 2.738903$), for which this over-representation was still significant for each of these controls.

A factor analysis of mixed data was performed on the numeric data which revealed that blood lead concentration and age had a moderate positive correlation, which the former moderately explained variation in the structure of the data and latter have a greater influence on variation. Blood lead concentration also had considerate correlations with urine albumin, serum creatinine, and waist circumference, which indicate that those who had higher lead exposure may have tended to be unhealthy metabolisms. It is noteworthy that while income poverty ratio greatly explained variation in the structure of the data, it was uncorrelated with blood lead concentration. For both the factor analysis and canonical correlation analysis, the data was normalized using z-score normalization.

These findings were further solidified through canonical correlation was performed to assess the relationship between social determinants of health (age, income poverty ratio, waist circumference) and metabolic markers (lead, creatinine, hemoglobin, eGFR). The results indicated that the main contributor to the first variation was age, which was associated with lower eGFR, higher creatinine, and higher blood lead concentrations. This points to associations between accumulated metabolic damage with age which could be a result of environmental toxin buildups.

To test the hypotheses that blood lead concentration was significantly correlated with age cohort membership, blood hemoglobin concentration being significantly explained by blood lead concentration, and mortality in the 2019 follow up by blood lead concentration, generalized linear models were specified with appropriate demographic controls and adjustment for survey design. The model for relationship between blood lead concentration and age cohort membership used the inverse gaussian family to approximate the greatly right skewed distribution of blood lead concentration. In that model, birth cohort, income-to-poverty ratio, waist circumference, and sex were significantly associated with blood lead concentration. Smoking status, diabetes status, and most race/ethnicity categories were not statistically significant. For the model for the relationship between blood hemoglobin concentration and blood lead concentration, hemoglobin concentration was significantly associated with income-to-poverty ratio, smoking status, waist circumference, sex, and the interaction between blood lead and birth cohort, but not to any of the other controls. In the model for the relationship between mortality and blood lead concentration with controls, none of the variables were significant at 95% confidence. However, when univariate versions of the models were tested all models were significant. Surprisingly, in all models tested, proportion of income to poverty ratio was not a significant predictor for lead exposure in any model tested.

Results

Survey-weighted GLM accounting for survey design were used to examine associations between blood lead levels, hemoglobin, and mortality, adjusting for sociodemographic and clinical covariates.

In the model for log blood lead levels, which used the inverse Gaussian family to adjust for the heavy skew, individuals born in 1978 and 1988 had significantly lower blood lead levels compared to the pre-1978 strata ($\beta = -0.316$, $SE = 0.019$, $p = .004$). Higher income-to-poverty ratio ($\beta = 0.040$, $SE = 0.004$, $p = .012$) and greater waist circumference ($\beta = 0.0028$, $SE = 0.0005$, $p = .031$) were associated with higher blood lead levels. Identifying as male was associated with lower blood lead levels ($\beta = -0.191$, $SE = 0.015$, $p = .006$). None other covariates were statistically significant.

In the model for hemoglobin, log blood lead levels were not independently associated with hemoglobin ($\beta = 8.35 \times 10^{-5}$, $SE = 9.36 \times 10^{-5}$, $p = .423$), but however, a significant interaction between blood lead and birth cohort was observed ($\beta = -3.26 \times 10^{-4}$, $SE = 9.94 \times 10^{-5}$, $p = .030$), indicating stronger negative associations in the younger cohort. Identifying as male and waist circumference were inversely associated with hemoglobin ($p < .001$ for both).

In the model for mortality, log blood lead was positively associated with mortality risk, though not statistically significant under survey weighting ($\beta = 0.221$, $SE = 0.031$, $p = .088$). Lower income-to-poverty ratio was associated with increased mortality risk ($\beta = -0.255$, $SE = 0.029$, $p = .073$). No other covariates showed consistent significant associations.

Discussion

Based on the results of this analysis, there is evidence to warrant further investigation that either old infrastructure or regulations noncompliance may have significantly contributed to mortality rates in older individuals. Adults showing high correlations with elevated blood lead concentrations measured between 2005-2006 had disproportionate mortality rates in the 2019 follow up regardless of their age. It is noteworthy that these results were consistent across socioeconomic groups which indicate that there is no clear evidence of these significantly being the result of environmental and municipal injustices which implies these effects could be due to widespread neglect.

Appendix

Table 1: Descriptive Statistics for Numeric Variables (Adult Cohort)

Variable	Mean	SD	Median	Min	Max	Skew	Kurtosis
Serum Creatinine (mg/dL)	-0.16	0.28	-0.22	-0.92	2.88	0.97	5.75
Urine Albumin (mg/L)	2.34	1.32	2.19	-1.56	9.56	0.96	2.56
Urine Creatinine (mg/dL)	4.70	0.69	4.81	1.61	6.52	-0.75	0.56
BMI (log)	3.18	0.29	3.19	2.46	4.87	0.14	-0.35
Waist Circumference (cm)	85.18	21.72	85.50	40.20	175.00	0.16	-0.37
Age (years)	28.00	24.08	19.00	0.00	85.00	0.80	-0.51
Income-to-Poverty Ratio	2.37	1.58	1.95	0.00	5.00	0.45	-1.15
C-Reactive Protein (mg/L)	0.37	0.80	0.12	0.01	17.50	7.80	98.90
eGFR	102.26	27.32	104.02	2.74	168.36	-0.44	-0.27
Blood Lead (μg/dL)	1.68	1.70	1.25	0.18	55.20	7.37	134.92
Hemoglobin (g/dL)	13.89	1.50	13.80	5.80	19.30	0.05	0.13
Exam Weight (2-year)	28180.99	28206.27	18037.12	0.00	156152.18	1.35	1.07

Table 2: Categorical Variable Distribution (Adult Cohort)

Variable	Distribution
Sex	Female: 5268; Male: 5080
Race/Ethnicity	White: 3928; Black: 2710; Mexican American: 2847; Other: 514; Other Hispanic: 349
Education Level	1: 1317; 2: 1616; 3: 2412; 4: 2934; 5: 2044; 7: 20; 9: 5
Diabetes Diagnosis	No: 9677; Yes: 554; Borderline: 108; Unknown: 9
Insulin Use	No: 10007; Yes: 310; Unknown: 31
Depression (PHQ-2)	0: 8110; 1: 1629; 2: 375; 3: 224; 9: 10
General Health	1: 1275; 2: 3101; 3: 4061; 4: 1657; 5: 253; 9: 1
Cancer Diagnosis	No: 9492; Yes: 850; Unknown: 6
Smoking (Recent)	No: 8112; Yes: 2234; Refused: 2
Mortality Status	No: 8385; Yes: 1963

Table 3: Descriptive Statistics for Numeric Variables (Born Pre-1978 Cohort)

Variable	Mean	SD	Median	Min	Max	Skew	Kurtosis
Serum Creatinine (mg/dL)	-0.09	0.29	-0.11	-0.92	2.88	1.23	7.66
Urine Albumin (mg/L)	2.28	1.38	2.12	-1.56	9.56	0.97	2.58
Urine Creatinine (mg/dL)	4.61	0.71	4.74	1.61	6.41	-0.73	0.37
BMI (log)	3.34	0.22	3.32	2.59	4.87	0.28	0.95
Waist Circumference (cm)	98.11	16.42	97.60	43.40	175.00	0.20	0.65
Age (years)	53.45	16.92	51.00	28.00	85.00	0.27	-1.08
Income-to-Poverty Ratio	2.76	1.59	2.50	0.00	5.00	0.16	-1.39
C-Reactive Protein (mg/L)	0.49	0.87	0.23	0.01	17.50	7.04	83.82
eGFR	88.83	23.57	90.00	2.74	157.39	-0.34	-0.01
Blood Lead (μg/dL)	2.02	1.71	1.59	0.18	26.40	4.11	32.41
Hemoglobin (g/dL)	14.21	1.57	14.30	5.80	19.30	-0.31	0.57
Exam Weight (2-year)	43075.19	32359.24	31860.55	0.00	156152.18	0.72	-0.63

Table 4: Categorical Variable Distribution (Born Pre-1978 Cohort)

Variable	Distribution
Sex	Female: 2097; Male: 2032
Race/Ethnicity	White: 2144; Black: 943; Mexican American: 761; Other: 168; Other Hispanic: 113
Education Level	1: 553; 2: 602; 3: 973; 4: 1110; 5: 882; 7: 6; 9: 3
Diabetes Diagnosis	No: 3537; Yes: 504; Borderline: 82; Unknown: 6
Insulin Use	No: 3939; Yes: 173; Unknown: 17
Depression (PHQ-2)	0: 3253; 1: 618; 2: 151; 3: 101; 9: 6
General Health	1: 387; 2: 1235; 3: 1583; 4: 783; 5: 141
Cancer Diagnosis	No: 3718; Yes: 407; Unknown: 4
Smoking (Recent)	No: 3127; Yes: 1002
Mortality Status	No: 3120; Yes: 1009

Table 5: Descriptive Statistics for Numeric Variables (Born 1978–1988 Cohort)

Variable	Mean	SD	Median	Min	Max	Skew	Kurtosis
Serum Creatinine (mg/dL)	-0.20	0.27	-0.22	-0.92	1.86	0.79	4.41
Urine Albumin (mg/L)	2.39	1.27	2.22	-1.56	9.38	0.97	2.52
Urine Creatinine (mg/dL)	4.76	0.66	4.84	1.79	6.52	-0.74	0.65
BMI (log)	3.08	0.28	3.05	2.46	4.33	0.52	-0.21
Waist Circumference (cm)	76.60	20.54	74.80	40.20	167.20	0.61	0.14
Age (years)	11.10	7.70	12.00	0.00	27.00	0.16	-1.06
Income-to-Poverty Ratio	2.10	1.52	1.65	0.00	5.00	0.66	-0.81
C-Reactive Protein (mg/L)	0.29	0.75	0.07	0.01	17.50	8.68	117.31
eGFR	111.18	25.96	113.28	7.31	168.36	-0.82	0.36
Blood Lead (μg/dL)	1.46	1.66	1.04	0.18	55.20	10.20	226.14
Hemoglobin (g/dL)	13.69	1.42	13.50	8.30	18.20	0.28	0.03
Exam Weight (2-year)	18292.24	19586.37	8624.87	0.00	144712.24	1.81	3.40

Table 6: Categorical Variable Distribution (Born 1978–1988 Cohort)

Variable	Distribution
Sex	Female: 3171; Male: 3048
Race/Ethnicity	Mexican American: 2086; Black: 1767; White: 1784; Other: 346; Other Hispanic: 236
Education Level	1: 764; 2: 1014; 3: 1439; 4: 1824; 5: 1162; 7: 14; 9: 2
Diabetes Diagnosis	No: 6140; Yes: 50; Borderline: 26; Unknown: 3
Insulin Use	No: 6068; Yes: 137; Unknown: 14
Depression (PHQ-2)	0: 4857; 1: 1011; 2: 224; 3: 123; 9: 4
General Health	1: 888; 2: 1866; 3: 2478; 4: 874; 5: 112; 9: 1
Cancer Diagnosis	No: 5774; Yes: 443; Unknown: 2
Smoking (Recent)	No: 4985; Yes: 1232; Refused: 2
Mortality Status	No: 5265; Yes: 954

Table 7: Variable Contributions to Principal Components (Factor Analysis)

Variable	Dim 1	Dim 2	Dim 3	Dim 4	Dim 5
Serum Creatinine (mg/dL)	5.74	3.59	16.11	3.31	0.37
Urine Albumin (mg/L)	0.00	7.11	1.02	1.01	16.71
Urine Creatinine (mg/dL)	0.10	2.92	0.07	16.39	19.27
BMI	9.71	0.47	18.68	0.97	2.47
Waist Circumference (cm)	11.74	0.33	16.23	1.50	1.24
C-Reactive Protein (mg/L)	0.84	0.97	2.46	1.77	1.91
Age (years)	17.95	0.19	0.35	1.31	0.51
Income-to-Poverty Ratio	2.08	11.27	0.73	0.00	0.21
Exam Weight (2-year)	7.56	16.09	0.15	0.01	0.00
Strata	1.33	2.94	5.07	0.25	12.59
Blood Lead (μg/dL)	1.19	3.37	4.04	0.41	2.34
Hemoglobin (g/dL)	2.21	0.70	0.03	25.41	6.82
eGFR	9.63	3.44	11.35	0.71	0.50

Table 8: Canonical Correlation Analysis Results

Function	**CV1**	**CV2**	**CV3**
Canonical Correlation (r_c)	0.545	0.240	0.129

Table 9: Standardized Canonical Coefficients

X Variables	**CV1**	**CV2**	**CV3**
income_poverty_ratio	-0.028	-0.445	0.910
age_years	-1.017	0.557	0.058
waist_circumference_cm	0.047	-1.033	-0.502
Y Variables	**CV1**	**CV2**	**CV3**
serum_creatinine_mg_dl	0.776	0.138	0.528
urine_albumin_mg_l	-0.076	0.285	-0.264
urine_creatinine_mg_dl	0.101	-0.452	-0.541
blood_lead_ug_dl	-0.148	0.530	-0.662
hemoglobin_g_dl	-0.445	-0.739	-0.238
egfr	1.380	0.126	0.038

Table 10: Structural Correlations (Canonical Loadings)

X Variables	**CV1**	**CV2**	**CV3**
income_poverty_ratio	-0.189	-0.436	0.880
age_years	-0.999	-0.024	-0.043
waist_circumference_cm	-0.457	-0.793	-0.402
Y Variables	**CV1**	**CV2**	**CV3**
serum_creatinine_mg_dl	-0.432	-0.020	0.229
urine_albumin_mg_l	-0.020	0.163	-0.448
urine_creatinine_mg_dl	0.193	-0.380	-0.613
blood_lead_ug_dl	-0.336	0.487	-0.585
hemoglobin_g_dl	-0.308	-0.735	-0.217
egfr	0.817	-0.137	-0.241

Table 11: Aggregate Redundancy Coefficients

Direction	**Variance Explained**
X | Y	0.144
Y | X	0.067

Table 12: Survey-Weighted Adjusted Model of Blood Lead Level

Predictor	Estimate (b)	SE	P Value
Intercept	0.5330	0.0754	.019
Born Pre-1978	-0.3158	0.0193	.004
Income-to-poverty ratio	0.0404	0.0045	.012
Smoking: Refused	0.0060	0.4294	.990
Smoking: Yes	-0.0721	0.0177	.055
Waist circumference (cm)	0.0028	0.0005	.031
Diabetes: No	0.0142	0.0392	.752
Diabetes: Unknown	-0.1034	0.0559	.206
Diabetes: Yes	-0.0003	0.0480	.996
Race: Mexican American	0.0311	0.0194	.251
Race: Other	0.0503	0.0497	.418
Race: Other Hispanic	0.0172	0.0436	.731
Race: White	0.0604	0.0275	.159
Male sex	-0.1913	0.0145	.006

Table 13: Survey-Weighted Model of Hemoglobin Level With Interaction

Predictor	Estimate (b)	SE	P Value
Intercept	0.0816	0.0008	$< .001$
Blood lead (µg/dL)	0.0000835	0.0000936	.423
Born Pre-1978	-0.0006105	0.0002575	.077
Income-to-poverty ratio	-0.0002758	0.0000713	.018
Smoking: Refused	-0.004041	0.001690	.075
Smoking: Yes	-0.002593	0.000177	$< .001$
Waist circumference (cm)	-0.0000644	0.0000040	$< .001$
Diabetes: No	-0.0003862	0.0005689	.535
Diabetes: Unknown	-0.002355	0.001639	.224
Diabetes: Yes	0.002177	0.000895	.072
Male sex	-0.005957	0.000239	$< .001$
Blood lead × Born Pre-1978	-0.0003264	0.0000994	.030

Table 14: Survey-Weighted Logistic Regression of Mortality With Covariate Adjustment

Predictor	Estimate (b)	SE	P Value
Intercept	0.2993	0.2243	.409
Blood lead (µg/dL)	0.2208	0.0306	.088
Born Pre-1978	0.3449	0.1414	.248
Income-to-poverty ratio	-0.2549	0.0292	.073
Smoking: Refused	-7.9755	0.7226	.058
Smoking: Yes	-0.2240	0.0710	.195
Waist circumference (cm)	-0.0115	0.0025	.136
Diabetes: No	-1.2099	0.1668	.087
Diabetes: Unknown	1.5236	1.2081	.427
Diabetes: Yes	0.1130	0.2189	.697
Race: Mexican American	-0.4491	0.0968	.135
Race: Other	-0.2522	0.1891	.410
Race: Other Hispanic	-1.1091	0.2112	.120
Race: White	0.4911	0.1103	.141
Male sex	0.0200	0.0681	.818

Figure 1: Pearson Correlations for Study Variates

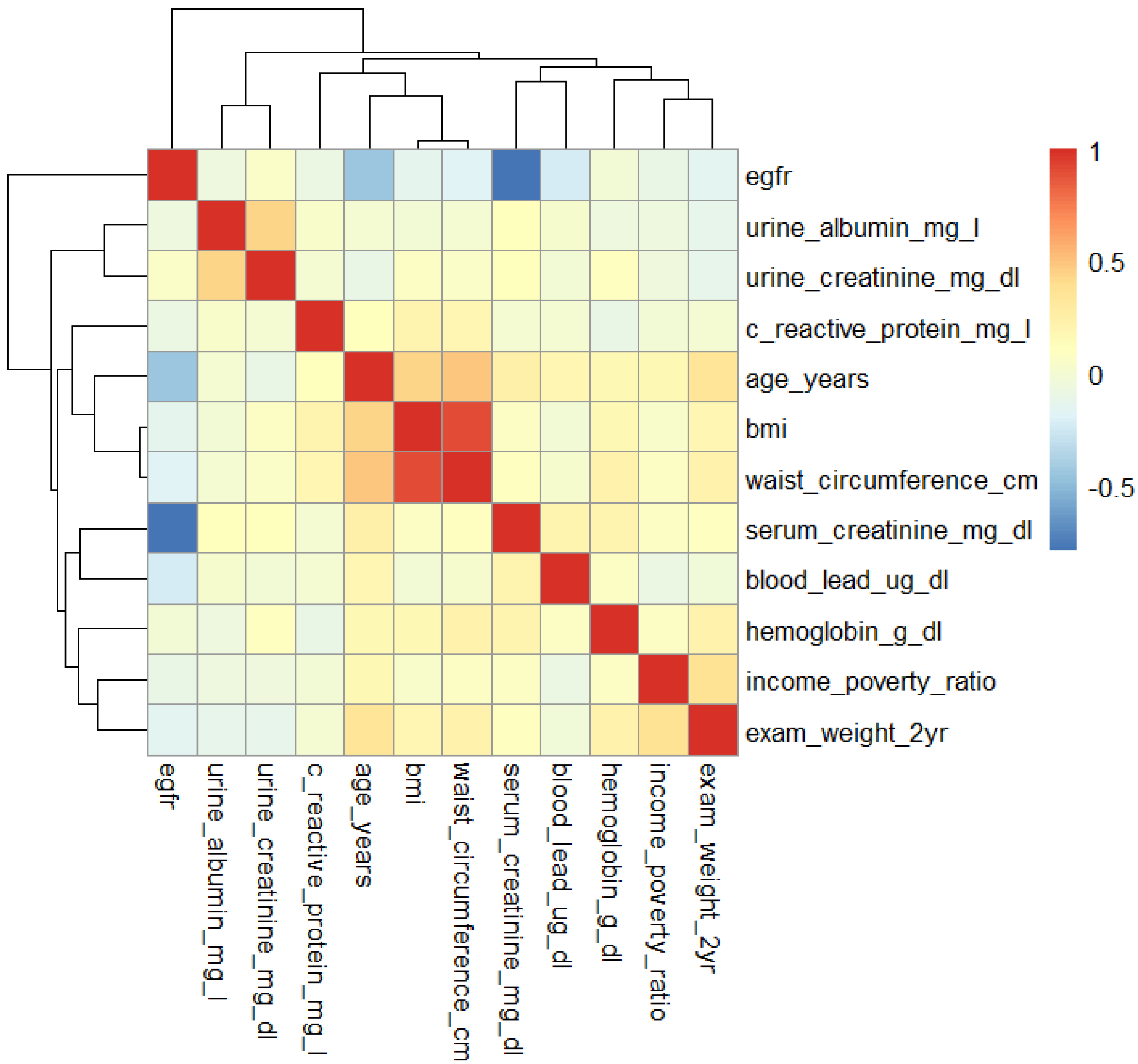

Figure 2: Violin Plots for Variables of Interests

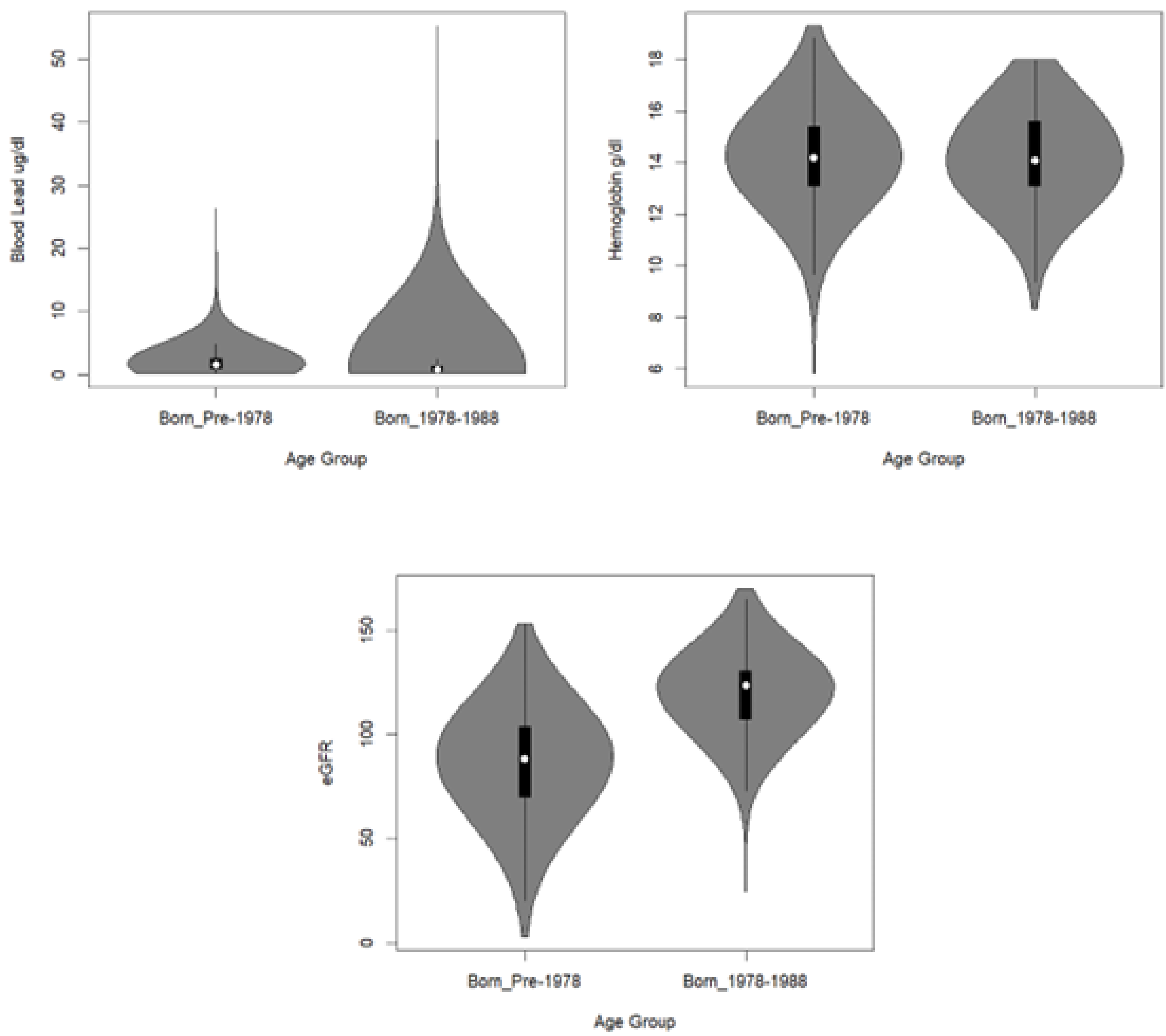

Figure 3: Bivariate Plots for Variables of Interests

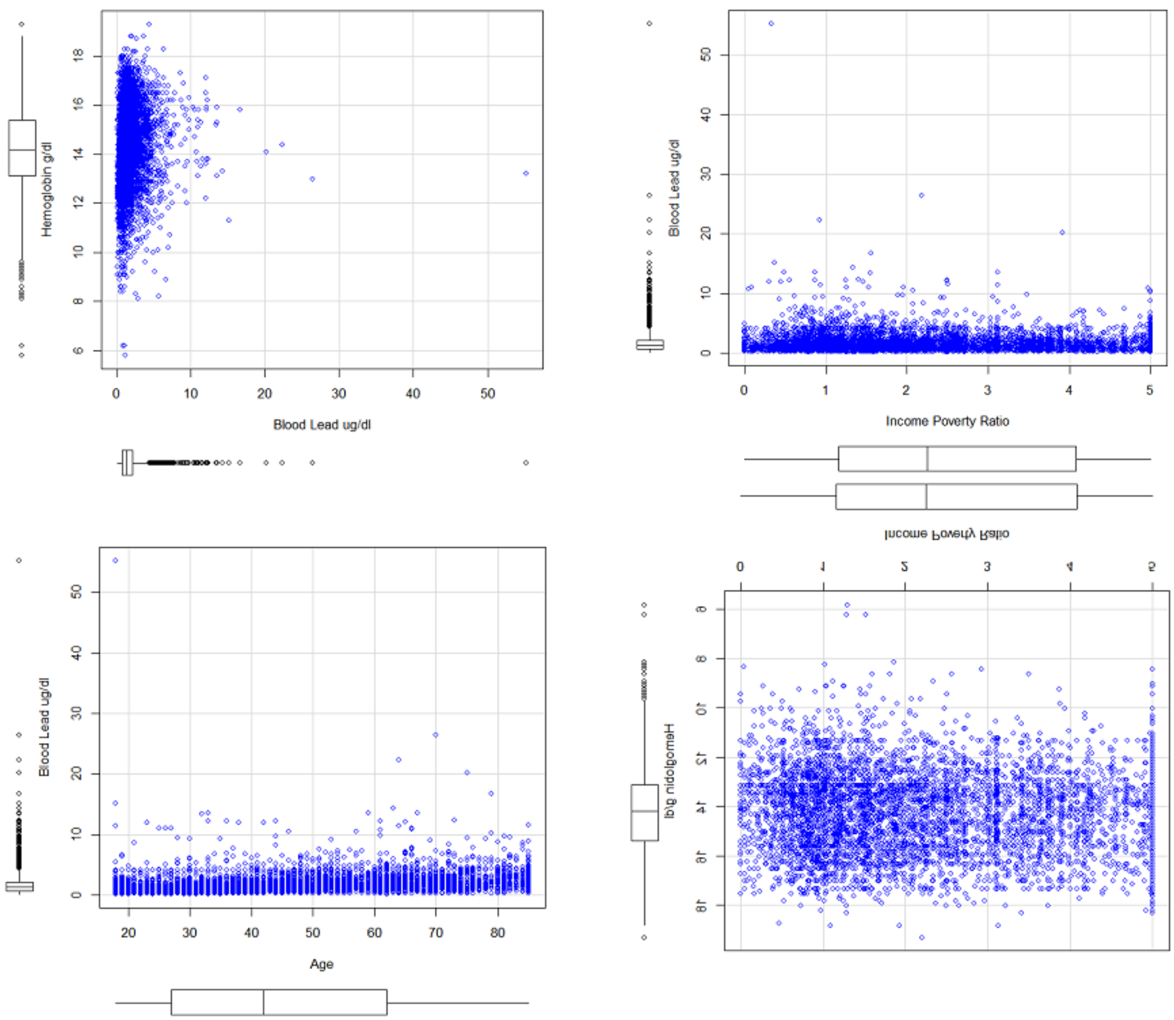

Figure 4: Distribution of Mortality Status by Elevated Blood Lead Levels

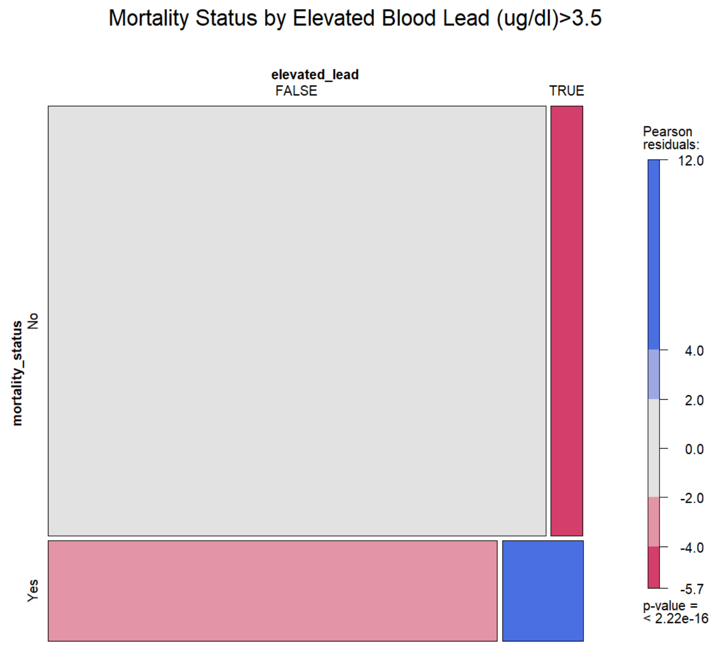

Figure 5: Canonical Correlation Helio Plot

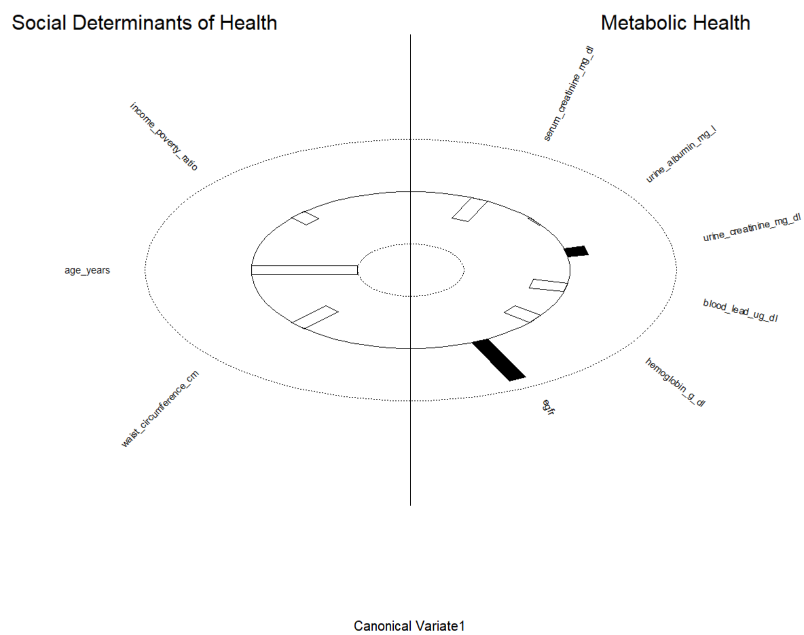

Figure 6: Factor Analysis Loading Plot